- Stomach pain, burning, or aching 1-4 hours after eating
- Use of antacids
- Feel hungry an hour or two after eating
- Heartburn when lying down or bending forward
- Temporary relief by using antacids, food, milk, or carbonated beverages
- Digestive problems subside with rest and relaxation
- Heartburn due to spicy foods, chocolate, citrus, peppers, alcohol, and caffeine

Further Assessment of Category V – Stomach

Further assessment is required in the following areas:

- Check for *H. pylori*
- Check for high aspirin use
- Check for stress-induced permeability
- Check for gastric cancer
- Check for autoimmunity:
  - Parietal cell antibodies
  - Intrinsic factor antibodies

MAF Blood Chemistry Key Association Category V – Stomach

- Check for hyperacidity:
  - Protein and globulin increased or decreased
  - BUN increased or decreased
  - Phosphorus decreased
- Check for possible *H. pylori* infection, which may alter acid production:
  - WBC increased or decreased
  - Neutrophils increased or decreased

MAF Nutritional Key Category V – Stomach*

*Nutritional Support for pH Balance, Healthy Stomach Acid Levels, and the Gastrointestinal System:**

1. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

   Alternative Formulas
   - HCL Pro-Zyme™◊ (Z26): 1-2 tablets with each meal.
   - SuperDigest-Zyme™◊ (Z22): 1-2 tablets with each meal.
2. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

*Nutritional Support for Gastrointestinal Health:**

1. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.
2. H-PLR™◊ (K32): 1-2 capsules with each meal.

*Nutritional Support for pH Balance and the Gastrointestinal System:**

1. H-PLR™◊ (K32): 1-2 capsules with each meal.
2. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

   Alternative Formulas
   - HCL Pro-Zyme (Z26): 1-2 tablets with each meal.
   - SuperDigest-Zyme (Z22): 1-2 tablets with each meal.
3. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.
4. Myco-Zyme™◊ (Z13): 1-2 capsules with meals.

*Nutritional Support for pH Balance and the Adrenals:**

1. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

Also Consider:

1. AdrenaCalm™◊ (KR16), (KR73): Apply in the morning, afternoon, and evening.

To support levels of hydrochloric acid and digestion, consider using EnzymixPro™◊ (K99).* Many individuals will notice supported digestion immediately when taking the enzyme they need to break down protein and prevent protein putrefaction.* Some of these individuals may, however, have increased issues with the ingestion of HCl. If an individual does not have a good response with EnzymixPro™◊ (K99), consider using Gastro-ULC™◊ (K29), which is a source of deglycyrrhizinated licorice, L-glutamine, flavonoids, bismuth, gamma oryzanol, chlorella, rhubarb, and mastic gum.* After 30 days, individuals can consider taking a small serving of HCl to support the stomach.* To support the bacterial environment in the stomach, H-PLR™◊ (K32) should be considered.* Cortisol from stress thins the lining of the stomach and makes an individual more susceptible to changes in the acidic environment.

Category VI – Pancreas Interpretation

Pancreatic insufficiency will cause symptoms similar to hypochlorhydria, but the symptoms are generally experienced after starch ingestion versus protein ingestion. Most people who have pancreatic insufficiency have it

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For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

secondary to hypochlorhydria. If the bolus does not become acidic enough once it leaves the stomach and enters the duodenum, it will not trigger the release of pancreatic enzymes and will hence induce secondary pancreatic insufficiency. Pancreatic insufficiency also commonly occurs from chronic gastrointestinal inoculation with a pathogen. In these cases, the pancreas is called into action to release enzymes to support the GI tract due to its anti-inflammatory and proteolytic nature. Primary pancreatic insufficiency is often due to diets high in starch and with individuals who have some type of blood sugar regulation disorder.

- Difficulty digesting roughage and fiber
- Indigestion and fullness last 2-4 hours after eating
- Pain, tenderness, soreness on left side under rib cage
- Excessive passage of gas
- Nausea and/or vomiting
- Stool undigested, foul smelling, mucus like, greasy, or poorly formed
- Frequent loss of appetite

Further Assessment of Category VI – Pancreas

Further assessment is required in the following areas:

- Check for excess-starch diet
- Check for failure of brain-gut axis
- Check for low thyroid
- Check for intestinal inflammation:
  - Check for infections
  - Check for gluten sensitivity/celiac disease
  - Check for intestinal permeability

MAF Blood Chemistry Key Association Category VI – Pancreas

- Pancreatic enzyme deficiency shows up many times as insulin resistance
- Check for pancreatic enzyme deficiency
- Check Category XI on Metabolic Assessment Form™ results to assess the peripheral utilization of sugars
- Usually only symptoms associated and not associated with blood chemistry markers

MAF Nutritional Key Category VI – Pancreas*

*Nutritional Support for the Pancreas:**

1. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.
2. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

*Nutritional Support for the Pancreas and Healthy Stomach Acid Levels:**

1. EnzymixPro™◊ (K99), (K99A): 1-2 capsules with each meal.

   Alternative Formula
   - HCL Pro-Zyme™◊ (Z26): 1-2 tablets with meals.

   Additional Formula
   - SuperDigest-Zyme™◊ (Z22): 1-2 tablets with meals.

*Dietary and Nutritional Support for the Small Intestine:**

1. ClearVite™◊ Basic Program: Three-week intestinal support program as directed by the ClearVite™◊ Basic Program Guide.
2. Gastro-ULC™◊ (K29): 1 tablet, 15-20 minutes before a meal, three times a day.

Category VII – Small Intestine Interpretation

The body intentionally keeps the bacterial amount low in the small intestine. There are several mechanisms that can explain why the concentration of bacteria is normally so low in the small intestine while fairly high (up to 10^9/mL) in the terminal ileum, which is located at the end of the small intestine. These include the following: many ingested bacteria are destroyed in the stomach (via acid) and the proximal small intestine (via bile and enzymes); intestinal peristaltic activities significantly limit the amount of bacteria on the mucosal layer by forcefully transporting the bacteria forward (this mechanism, in addition to the ileocecal valve, prevents the bacteria from moving from the colon back to the ileum); and the immune system and secretory IgA (sIgA) play a role in selectively preventing the proliferation of certain bacteria. However, these mechanisms may function at suboptimal levels for multiple reasons, including: stress; a weakened immune system; lowered gastric acid—for instance, in those who regularly consume over-the-counter antacids, in the elderly, etc.; and irregular peristalsis—peristalsis can be affected in those who, for example, have stressful lifestyles or bad sleeping patterns. These irregular muscle movements may present with either decreased or increased bowel frequency, and/or a combination of both. These factors can increase the chance for a higher bacterial presence in the small intestine, which can cause digestive discomfort due to fermentation and gas formation, especially after ingestion of sugars, fiber, certain probiotics, and starch. If left unchecked, the altered environment of the small intestine can affect the absorption of essential nutrients, such as fats and proteins.

* This statement has not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease.

Apex Energetics™ products are intended for nutritional use and health maintenance only and should not replace or delay the use of any medication.

- Abdominal distention after consumption of fiber, starches, and sugar
- Abdominal distention after certain probiotic or natural supplements
- Decreased gastrointestinal motility, constipation
- Increased gastrointestinal motility, diarrhea
- Alternating constipation and diarrhea
- Suspicion of nutritional malabsorption
- Frequent use of antacid medication
- Diagnosed with celiac disease, irritable bowel syndrome, diverticulosis/diverticulitis, or leaky gut syndrome

Further Assessment of Category VII – Small Intestine

Further assessment is required in the following areas:

- Check for failure of brain-gut axis
- Check for gastric acid activity
- Check for history of acid blocking medications
- Check for gut motility:
  - Autonomic and enteric nervous system
- Check for alcohol consumption
- Check for intestinal inflammation:
  - Check for infections
  - Check for gluten sensitivity/celiac disease
  - Check for IBS
  - Check for Crohn's disease
- Intestinal Permeability
  - Cyrex® Laboratories Array 2
- Gluten Sensitivity/Crohn's Disease
  - Cyrex® Laboratories Array 3 Wheat/Gluten Proteome Reactivity & Autoimmunity
- IBS/Food Reactivity
  - Cyrex® Laboratories Array 10 Multiple Food Immune Reactivity Screen

MAF Blood Chemistry Key Association Category VII – Small Intestine

Usually only symptoms associated and not associated with blood chemistry markers. However, a CBC and stool exam may help recognize infections, anemia, etc.

MAF Nutritional Key Category VII – Small Intestine*

*Dietary and Nutritional Support for the Small Intestine:**

1. ClearVite-GL™◊ (K95), (K118): 1 scoop, with up to 4-6 ounces of water, once a day. However, a CBC and stool exam may help recognize infections, anemia, etc.
2. Sibiotica™◊ (K97): 1 capsule once a day.
3. RepairVite-SE™◊ (K98): 1 scoop, with up to 4-6 ounces of water, once a day.
4. EnzymixPro™◊ (K99), (K99A): 1-2 capsules once a day.
5. EnteroVite™◊ (K100): 1 capsule 3 times a day.

   Alternative Formula
   - Tributyrin 350™◊ Active (K132): 1 softgel, 1-2 times a day, with a meal.
6. Dietary restrictions as directed by the Small Intestinal Microbial Environment Dietary and Nutritional Guide.
7. Exercises to increase motility as recommended by your healthcare professional.

Category VIII – Biliary Interpretation

Biliary insufficiency or stasis is a condition in which the body does not make enough bile or the bile is thick. Bile is a complex fluid consisting of various components including bile salts. It is secreted by the liver, and it is temporarily stored in the gallbladder to aid in fat digestion. The hormone cholecystokinin, released during eating, triggers the contraction of the gallbladder, which expels stored bile into the small intestine. In the small intestine, bile emulsifies fat, permitting its digestion by pancreatic enzymes.
The absorption of fat-soluble vitamins also requires bile emulsification. Biliary insufficiency or stasis can be caused by diets high in fat, elevated estrogen, or may be secondary to hypochlorhydria. Also note that if the individual has had his/her gallbladder removed, he/she will complain of biliary insufficiency symptoms because his/her cystic duct, which is very small, is now taking on the role of the gallbladder.

- Greasy or high-fat foods cause distress
- Lower bowel gas and/or bloating several hours after eating
- Bitter metallic taste in mouth, especially in the morning
- Burpy, fishy taste after consuming fish oils
- Unexplained itchy skin
- Yellowish cast to eyes
- Stool color alternates from clay colored to normal brown
- Reddened skin, especially palms
- Dry or flaky skin and/or hair
- History of gallbladder attacks or stones
- Gallbladder removed

Further Assessment of Category VIII – Biliary

Further assessment is required in the following areas:

- Check for high-fat diet with an excess of fried foods
- Check for failure of brain-gut axis
- Check for low thyroid

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

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- Check for excess estrogens:
  - Exogenous (oral contraceptives, hormone replacement therapy, etc.)
  - Endogenous

MAF Blood Chemistry Key Association Category VIII – Biliary

- GGTP, alkaline phosphatase—don't usually show up until obstruction
- MAF symptoms may be due to thick bile, pre-gallstones, or gallstones
- Some individuals may have no symptoms because they avoid foods that cause them

MAF Nutritional Key Category VIII – Biliary*

*Nutritional Support for the Biliary System:**

1. BileMin™◊ (K11): 1-2 capsules with meals.
2. Lypomin-LV™◊ (K33): 1-3 capsules with meals.
   Alternative Formula
   - Metacrin-DX™◊ (K10): 1-2 capsules 2-3 times a day.
3. Super Methyl-SP™◊ (K112): 1-2 capsules twice a day.
   Alternative Formula
   - Methyl-SP™◊ (K14): 1-2 capsules with meals.

Additional Support Considerations:

1. C Liquazome™◊ (K139), (K139A): 1 teaspoon (4.93 mL or 1 packet) 1-2 times a day.
   Alternative Formula
   - C-Zyme 1000™◊ (Z4): 1 tablet with meals.
2. OmegaCo3-SE™◊ (K113): 2 softgels 1-2 times a day.
   Alternative Formulas
   - OmegaCo3™◊ (K7): 1 tablespoon 1-2 times a day.
   - Super EFA Complex™◊ (K8): 1-2 tablespoons with meals.§
3. Mg/K-Zyme™◊ (Z11): 1 tablet with meals.

§Currently unavailable

Category IX – Hepatic Detoxication Interpretation

This category screens for symptoms related to hepatic detoxification integrity, more appropriately known as "hepatic biotransformation" or "hepatic detoxication." Many diverse symptoms may occur if the liver is inefficient in its abilities to convert compounds with phase I and phase II clearance pathways.

- Acne and unhealthy skin
- Excessive hair loss
- Overall sense of bloating
- Bodily swelling for no reason
- Hormone imbalances
- Weight gain
- Poor bowel function
- Excessively foul-smelling sweat

Further Assessment of Category IX – Hepatic Detoxication

Further assessment is required in the following areas:

- Check for excess alcohol, drugs, and environmental pollutants
- Check for hormone overload
- Check for poor diet

MAF Blood Chemistry Key Association Category IX – Hepatic Detoxication

- Usually only symptoms associated and not associated with blood chemistry markers
- May be suggested from functional elevation of SGOT, SGPT, and/or GGTP

MAF Nutritional Key Category IX – Hepatic Detoxication*

*Dietary and Nutritional Support for Hepatic Biotransformation:**

1. ClearVite™◊ Formula (K84, 95, 105, 108, 118, or 142): As directed by the ClearVite™◊ Basic Program Guide.
2. Hepato-Synergy™◊ (K69): 1 packet once a day or as needed.
3. Trizomal™◊ Glutathione (K122), (K129A): 5 mL (approx. 1 teaspoon or 1 packet) 1-2 times a day.
4. Nourish Greens™◊ (K67): 1 scoop, with up to 8 ounces of water, once a day.

Category X – Sugar Metabolism Interpretation

(Also check for Hypoglycemia)

Reactive hypoglycemia refers to a state of low blood sugar, or states in which blood sugar levels spike up and down. Hypoglycemics suffer from fatigue and low energy because they do not supply their bodies with the amount and/or frequency of energy needed to support their metabolic system. Hypoglycemia is usually accompanied by adrenal exhaustion because the adrenals are called into play when

blood sugar levels spike or are low. The adrenals eventually become exhausted and need support. In individuals with hypoglycemia, dietary and lifestyle changes need to be made with regard to not skipping breakfast and snacking with low glycemic foods during the day.

- Crave sweets during the day
- Irritable if meals are missed
- Depend on coffee to keep going/get started
- Get light-headed if meals are missed
- Eating relieves fatigue
- Feel shaky, jittery, or have tremors
- Agitated, easily upset, nervous
- Poor memory, forgetful between meals
- Blurred vision

Further Assessment of Category X – Sugar Metabolism

Further assessment is required in the following areas:

- Check for diet:
  - Missing meals
  - Sugar, coffee, or fruit for meals
- Check for functional adrenal imbalance

MAF Blood Chemistry Key Association Category X – Sugar Metabolism

- Check for hypoglycemia:
  - Decreased glucose levels
  - Decreased LDH levels

MAF Nutritional Key Category X – Sugar Metabolism*

*Nutritional Support for Sugar Metabolism:**

1. Proglyco-SP™◊ (K13): 1 capsule 3 times a day, or as needed, and/or GlycemoVite™◊ (K85): 1 scoop, with up to 8 ounces of water, between meals.
2. Butyric-FBR™◊ (K133): 2 capsules once a day.

   Alternative Formula
   - Fibromin™◊ (K25): 1 capsule, 2 times a day, or as needed.

Additional Support Considerations:

1. Adaptocrine™◊ (K2), (K124): 1-3 capsules with meals.
2. OmegaCo3-SE™◊ (K113): 2 softgels 1-2 times a day.

   Alternative Formulas
   - OmegaCo3™◊ (K7): 1 tablespoon 1-2 times a day.
   - Super EFA Complex™◊ (K8): 2 tablespoons once a day.§

Also Consider:

1. AdrenaStim™◊ (KR15), (KR72): Apply in the morning and at noon.

§Currently unavailable

Nutritional support for sugar metabolism should include:

Proglyco-SP™◊ (K13), which supplies the essential nutrients, minerals, and glandulars to support the adrenals and blood sugar metabolism; Adaptocrine™◊ (K2), (K124), which contains herbs that support sugar metabolism and the body's mental functionality under temporary stress; and Super EFA Complex™◊ (K8), which offers a source of essential fatty acids.*

Category XI – Peripheral Utilization of Sugars Interpretation

(Also check for Insulin Resistance)

Insulin resistance refers to a physiological pattern in which insulin receptor sites no longer respond to insulin. Insulin is a hormone that is secreted by the pancreas when serum glucose levels are elevated after meals. Insulin's job is to stimulate receptor sites, which will then transport glucose into the cells.

- Insulin resistance is caused by diets and physiological changes that cause prolonged and frequent release of insulin. Once insulin is repeatedly exposed to receptor sites, the sites will shut down and become nonresponsive. Insulin resistance will then lead to elevations of serum glucose because it cannot be carried into the cell, which leads to alterations in cholesterol and fat metabolism. Insulin resistance places great demands on the beta cells of the pancreas for insulin output, and may eventually lead to diabetes if this is not addressed properly.
- Fatigue after meals
- Crave sweets during the day
- Eating sweets does not relieve cravings for sugar
- Must have sweets after meals
- Waist girth is equal or larger than hip girth
- Frequent urination
- Increased thirst and appetite
- Difficulty losing weight

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For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.